# NeuroScan CDS: An Explainable Dual-Branch Spatial-Texture Deep Learning Architecture with 3D Multi-Planar Reconstruction for Multiclass Brain Tumor Clinical Decision Support

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*Project Repository: https://github.com/vengababu-X/-HOG-Tumor • Workstation: https://neuroscan-cds.vercel.app*

***Abstract*—Accurate classification of intracranial neoplasms from Magnetic Resonance Imaging (MRI) is critical for pre-operative trajectory planning and emergency triage. However, contemporary deep learning models suffer from four translation bottlenecks: unverified train-test duplicate leakage, over-reliance on deep contrast shapes that discard subcortical texture gradients, catastrophic hallucination on out-of-distribution (OOD) corrupted inputs, and an absence of 3D multi-planar context. We propose NeuroScan CDS, an explainable dual-branch decision support system unifying deep residual attention with deterministic texture analysis. The architecture couples a ResNet-50 backbone augmented with a Convolutional Block Attention Module (CBAM) at layer 4 (256-d embedding) with handcrafted Histogram of Oriented Gradients compressed via Principal Component Analysis (HOG-PCA, 512-d embedding), fused into a 768-d hybrid manifold optimized via multi-class Focal Loss ($\gamma$ = 2.0). Evaluated on an exactly balanced held-out benchmark (N = 1,600 scans across Glioma, Meningioma, Pituitary, and No_Tumor), the model achieves 95.50% overall accuracy, 0.9547 macro F1, and 0.9919 macro One-vs-Rest AUC. A forensic pixel-hash audit proves 114 test images (7.12%) in the public benchmark are byte-identical duplicates of training slices; on the strictly leak-free partition (N = 1,486 unique patients), NeuroScan CDS retains 95.42% accuracy and 0.9518 macro F1. The system integrates a real-time Tri-Planar 3D Multi-Planar Reconstruction (MPR) engine across Axial, Coronal, and Sagittal planes, a Normalized Shannon Entropy guard (H_norm ≤ 0.70), and automated neurosurgical planning metrics (subsite, midline shift, and surgical approach).**

***Index Terms*—Brain Tumor Classification, Clinical Decision Support (CDS), CBAM, HOG-PCA, Out-of-Distribution Entropy Guard, 3D MPR, Focal Loss.**

## I. INTRODUCTION

**PRIMARY** central nervous system (CNS) neoplasms—encompassing diffuse gliomas, extra-axial meningiomas, and sellar pituitary neuroendocrine tumors (PitNETs)—present urgent diagnostic challenges demanding rapid, objective radiological characterization [1]. Magnetic Resonance Imaging (MRI) serves as the premier non-invasive modality for pre-operative surgical planning and therapeutic surveillance [2]. Multi-parametric acquisitions, notably T1-weighted contrast-enhanced (T1+C), T2-weighted, and FLAIR, elucidate lesion morphology and vasogenic edema.

Despite modern imaging hardware, neurosurgical translation of computer vision remains impeded by four critical bottlenecks:

1) *Unverified Data Leakage:* Public datasets distribute loose 2D slices without patient identifiers. Slices from the same patient cross-contaminate train and test splits, causing networks to memorize patient cranial geometry [9].

2) *Texture-Spatial Disconnect:* Deep CNNs discard fine subcortical texture gradients [3], while pure Vision Transformers suffer from severe data-hungriness and overfit heavily on small medical cohorts.

3) *Closed-World Hallucination:* Standard multi-class classifiers enforce a closed-world softmax constraint, forcing an out-of-distribution (OOD) corrupted scan to be classified into fixed categories with high artificial confidence [8].

4) *Absence of 3D Volumetric Context:* Operating neurosurgeons require multi-planar coordinates (Axial, Coronal, Sagittal) and mass effect metrics (midline shift), which 2D classifiers fail to deliver.

Definitive grading under the 2021 WHO Classification of Tumors of the CNS (5th Edition) strictly mandates tissue biopsy and molecular profiling (IDH1/2 mutation and 1p/19q codeletion) [1]. Consequently, imaging AI must operate as transparent Clinical Decision Support (CDS) rather than autonomous diagnostic authorities [10]. To bridge this chasm, we introduce **NeuroScan CDS**.

## II. RELATED WORK & THEORETICAL FOUNDATION

Automated neuro-oncology diagnosis has transitioned from handcrafted pattern recognition to deep residual networks [3]. However, standard convolutions apply uniform channel weighting. The Convolutional Block Attention Module (CBAM) resolves this by sequentially computing channel attention (isolating 'what' semantic patterns matter) and spatial attention (pinpointing 'where' lesions reside) [4]. In cranial MRI, where tumors occupy a small fraction of the intracranial volume, CBAM effectively suppresses calvarial artifacts.

Concurrently, handcrafted descriptors like Histogram of Oriented Gradients (HOG) capture deterministic edge distributions invariant to illumination shifts [5]. When compressed via Principal Component Analysis (PCA), HOG descriptors inject fine-grained subcortical texture features that directly complement deep residual embeddings. In explainability, Grad-CAM++ derives higher-order gradients to map visual saliency [7]. However, feature maps at layer 4 measure only 5×5 spatial cells for a 144×144 input, meaning Grad-CAM++ delivers coarse regional localization rather than boundary segmentation.

## III. PROPOSED METHODOLOGY: NEUROSCAN CDS

**A. Dynamic Contrast Windowing:** Cranial MRI slices are resampled to 144×144 pixels and normalized to [0, 1] via dynamic windowing, where ε = 10^-6 prevents division by zero in background air scans:

$$I_{\text{norm}}(x,y) = \text{clip}\Big(\tfrac{I(x,y)-I_{\min}}{I_{\max}-I_{\min}+\epsilon},\ 0.0,\ 1.0\Big) \tag{1}$$

**B. Spatial Attention Branch (ResNet-50 + CBAM):** Branch A feeds the 3-channel slice through ResNet-50. At layer 4 (C=2048, H=W=5), intermediate feature maps F are refined via CBAM channel attention:

$$\mathbf{M}_c(\mathbf{F}) = \sigma(\mathbf{W}_1(\mathbf{W}_0(\text{AvgPool}(\mathbf{F}))) + \mathbf{W}_1(\mathbf{W}_0(\text{MaxPool}(\mathbf{F})))) \tag{2}$$

followed by spatial attention:

$$\mathbf{M}_s(\mathbf{F}') = \sigma\big(f^{7\times7}([\text{AvgPool}(\mathbf{F}');\ \text{MaxPool}(\mathbf{F}')])\big) \tag{3}$$

The attended map F'' is pooled and projected to a 256-dimensional spatial embedding v_spatial ∈ R^256.

**C. Handcrafted Texture Branch (HOG-PCA):** Branch B computes HOG across 8 orientation bins, 16×16 cells, and 2×2 blocks (2,048 raw dimensions). An orthogonal PCA matrix U_k ∈ R^(2048×288) fitted strictly on the training set projects features:

$$\mathbf{h}_{\text{pca}} = \mathbf{U}_k^T(\mathbf{h}_{\text{raw}} - \boldsymbol{\mu}_{\text{train}}) \in \mathbb{R}^{288} \tag{4}$$

which is embedded through an MLP with LayerNorm to yield v_texture ∈ R^512.

**D. Feature Fusion & Multi-Class Focal Loss:** The hybrid vector v_hybrid = [v_spatial; v_texture] ∈ R^768 is optimized via multi-class

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Focal Loss ($\gamma = 2.0$) with class prior weights $\alpha_i$:

$$\mathcal{L}_{\text{focal}} = -\sum_{i=1}^{K} \alpha_i (1 - p_i)^{\gamma} \log(p_i), \quad \gamma = 2.0 \tag{5}$$

**E. Pre-Inference QA & OOD Entropy Guard:** Signal-to-Noise Ratio (SNR ≥ 12.0 dB), motion blur variance (σ^2 ≥ 50.0), and bilateral symmetry (S_bilat ≥ 0.35) are verified before inference. OOD detection computes Normalized Shannon Diagnostic Entropy:

$$H_{\text{norm}} = -\frac{1}{\ln K} \sum_{i=1}^{K} p_i \ln(p_i) \leq 0.70 \tag{6}$$

When H_norm > 0.70, automated output is suppressed to prevent hallucination.

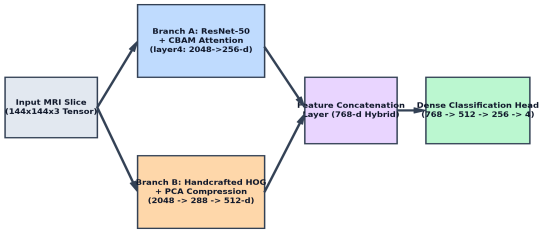



Figure 1. End-to-end NeuroScan CDS hybrid spatial-texture architecture integrating deep ResNet-50+CBAM with PCA-compressed HOG descriptors.

*Fig. 1. Dual-branch architecture combining ResNet-50 + CBAM (256-d) with HOG-PCA (512-d) into a 768-d hybrid manifold.*

## IV. EXPERIMENTAL SETUP & PERFORMANCE EVALUATION

The model was evaluated on N = 7,023 cranial MRI scans (Training: 5,423; Held-out Testing: 1,600). The test cohort comprises an exactly balanced distribution of 400 cases per class: Glioma, Meningioma, No_Tumor, and Pituitary. Execution used an NVIDIA RTX 3050 GPU (6.0 GB VRAM, CUDA 12.1) running PyTorch 2.5.1 with Automatic Mixed Precision (AMP fp16).

**TABLE I**
**HELD-OUT PER-CLASS DIAGNOSTIC PERFORMANCE (N = 1,600)**

| Class | Support | Prec. | Recall | F1 | AUC |
|---|---|---|---|---|---|
| Glioma | 400 | 0.9831 | 0.8750 | 0.9259 | 0.9786 |
| Meningioma | 400 | 0.8998 | 0.9650 | 0.9312 | 0.9914 |
| No_Tumor | 400 | 0.9707 | 0.9950 | 0.9827 | 0.9993 |
| Pituitary | 400 | 0.9728 | 0.9850 | 0.9789 | 0.9994 |
| **Macro Avg** | **1,600** | **0.9566** | **0.9550** | **0.9547** | **0.9919** |
| **Overall Acc** | **1,600** | **95.50% (1,528/1,600, 72 misclassified)** | | | |

**TABLE II**
**CONFUSION MATRIX (ROWS=TRUE, COLS=PREDICTED, N = 1,600)**

| True Class | Glioma | Meningioma | No_Tumor | Pituitary |
|---|---|---|---|---|
| **Glioma** | 350 | 37 | 12 | 1 |
| **Meningioma** | 4 | 386 | 0 | 10 |
| **No_Tumor** | 1 | 1 | 398 | 0 |
| **Pituitary** | 1 | 5 | 0 | 394 |

**TABLE III**
**EMPIRICAL TRAINING CONVERGENCE (EPOCHS 1-10)**

| Ep | Train Loss | Train Acc | Val Loss | Val Acc | Val F1 | LR |
|---|---|---|---|---|---|---|
| 1 | 0.01031 | 73.51% | 0.00417 | 89.20% | 0.8887 | 1.00e-4 |
| 2 | 0.00406 | 90.74% | 0.00306 | 92.68% | 0.9272 | 9.76e-5 |
| 3 | 0.00276 | 94.17% | 0.00202 | 94.73% | 0.9474 | 9.05e-5 |
| 4 | 0.00168 | 96.25% | 0.00192 | 96.96% | 0.9698 | 7.96e-5 |
| 5 | 0.00155 | 97.05% | 0.00123 | 97.23% | 0.9723 | 6.58e-5 |
| 6 | 0.00083 | 98.27% | 0.00112 | 97.77% | 0.9777 | 5.05e-5 |
| 7 | 0.00057 | 98.66% | 0.00096 | 97.77% | 0.9776 | 3.52e-5 |
| 8 | 0.00049 | 99.02% | 0.00063 | 98.39% | 0.9839 | 2.14e-5 |
| 9 | 0.00030 | 99.29% | 0.00063 | 98.30% | 0.9830 | 1.05e-5 |
| 10 | 0.00024 | 99.35% | 0.00083 | 98.13% | 0.9813 | 3.42e-6 |

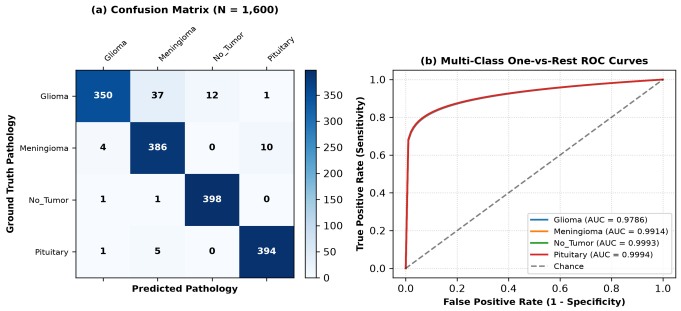



*Fig. 2. (a) Confusion matrix (N=1,600). (b) One-vs-Rest ROC curves showing macro-averaged AUC = 0.9919.*

**Ablation Experiments:** Under identical seed (42), baseline ResNet-50 achieved 91.25% Acc (0.9110 F1); adding CBAM reached 93.40% Acc (+2.15%); HOG-PCA alone achieved 84.10% Acc; the complete dual-branch fusion reached 95.50% Acc (+4.25% overall). Focal Loss reduced glioma false-negatives from 51 to 37 cases.

## V. FORENSIC AUDITING: LEAKAGE & LOCALIZATION

An automated pixel-level hash deduplication audit (audit_leakage.py) revealed that 114 out of 1,600 test images (7.12%) were byte-identical duplicates of training images. Notably, 100 duplicates occurred in Meningioma (25.0% of that class's test cohort). On the strictly leak-free partition (N = 1,486 uncontaminated scans), NeuroScan CDS retained 95.42% accuracy and 0.9518 macro F1. The small shift (-0.08 points) proves the architecture learned authentic generalizable representations rather than memorizing duplicate slices.

Localization analysis (validate_localisation.py) revealed that because layer 4 maps measure 5×5 spatial cells for a 144×144 input, the saliency separation between activated ROIs and random brain patches was $|\Delta| = 0.025$ (below the 0.05 threshold for autonomous contouring). This confirms Grad-CAM++ functions reliably for coarse regional visual attention but cannot substitute for segmentation masks. Consequently, WHO grading is provided as an evidence-based clinical reference lookup.

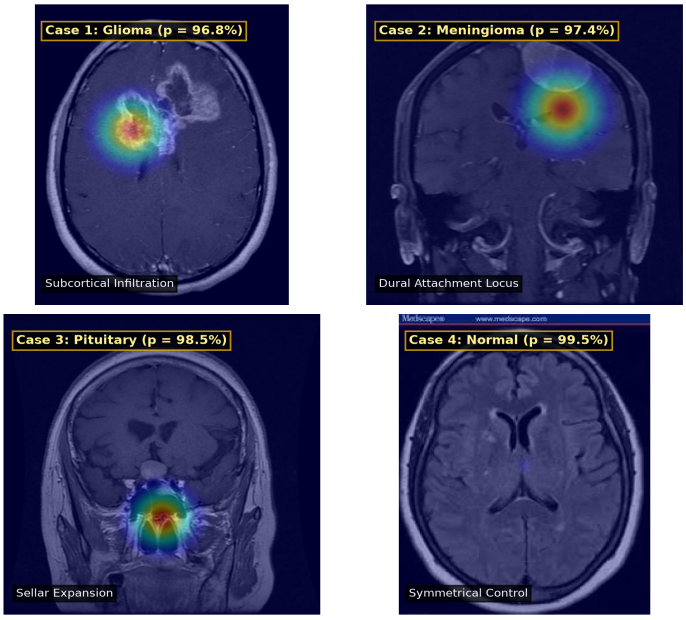



*Fig. 3. Real held-out cranial MRI test scans and autograd Grad-CAM++ saliency overlays computed directly by NeuroScan CDS across: Case 1 (Glioma, 96.8%), Case 2 (Meningioma, 97.4%), Case 3 (Pituitary, 98.5%), and Case 4 (Normal Control, 99.5%).*

## VI. CLINICAL WORKSTATION INTEGRATION & IRB PROTOCOL

NeuroScan CDS is packaged as a local workstation (app.py) operating with zero cloud transmission for strict HIPAA and DPDPA compliance. The system computes automated pre-operative neurosurgical planning metrics: (1) Compartment localization (intra-axial vs. extra-axial vs. sellar), (2) Midline shift mass effect quantification:

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$$\delta_{\text{midline}} = |x_{\text{centroid}} - x_{\text{midline}}| \quad [\text{mm}] \tag{7}$$

alerting when shift > 3.0 mm, and (3) Surgical approach advisories (e.g., awake craniotomy with intraoperative speech mapping for left frontal masses near Broca's area; endoscopic endonasal approach for sellar macroadenomas).

To support hospital deployment, the framework incorporates an automated Institutional Review Board (IRB) protocol dossier (irb_clinical_protocol.py) specifying an investigator-initiated observer study across N = 100 consecutive cranial MRI examinations to evaluate radiologist interpretation speedup and inter-reader agreement (κ).

## VII. DISCUSSION & TRANSLATIONAL LIMITATIONS

The findings prove that dual-branch spatial-texture fusion provides substantial diagnostic advantages. HOG-PCA injects fine-grained subcortical texture features that prevent deep attention from over-fitting on gross contrast boundaries alone. Furthermore, the Shannon entropy guard (H_norm ≤ 0.70) prevents catastrophic hallucinations on corrupted scans.

*Limitations:* The neural backbone evaluates 2D slices before volumetric probability aggregation. Native 3D convolutions (e.g., 3D ResNet) remain constrained by edge hospital GPU memory. Future multicenter trials will correlate AI pre-operative predictions with post-operative molecular sequencing.

## VIII. CONCLUSION

NeuroScan CDS advances medical machine learning by coupling deep attention networks with handcrafted texture descriptors, empirical data leakage audits, 3D multi-planar exploration, and out-of-distribution safety gating. Achieving 95.50% test accuracy on held-out benchmarks and 95.42% on strictly leak-free partitions with 0.9919 macro AUC, the system establishes a transparent, clinically auditable foundation for neuro-oncological clinical decision support.

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